

User Manual

Vanguard® Breast MRI Tabletop Coil 3.0T, 127.73 MHz

For use with 3.0T GE MRI Systems With P Ports

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Preface

Regulatory Notification Information



Federal Law restricts this device to sale by or on the order of a physician.

The Tabletop system must be installed and put into service according to the EMC information provided in this document. The Tabletop should be used only in a shielded location.

Portable and mobile RF communications equipment can affect the Tabletop system.

Using accessories, cables and transducers other than those designed for the Tabletop may significantly degrade electromagnetic emissions and immunity performance. Do not use accessories, transducers and cables other than those specified in this manual.

About This Guide

This manual describes the functions, features, use, safety precautions, and maintenance for the Sentinelle Tabletop Breast MRI Tabletop with 2/10/16-Channel Coil Array for GE 3.0T MRI System with P Ports, referred to in this guide as the Tabletop. To use the Tabletop safely and effectively, it is essential that you review and familiarize yourself with this manual in its entirety before using the system.

The Tabletop is also available in a 10-channel configuration or a 16-channel imaging-only configuration.

For an imaging-only configuration, sections that involve intervention or biopsy related items are not applicable.

NOTE: To upgrade to a complete system, contact your Sentinelle Medical sales representative.

The graphics, figures, and medical images used in this manual are examples only. The display and design may vary on your system.

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

Symbols and Safety Notices

The following symbols are used on Sentinelle Medical products to highlight information which, if not correctly followed, will directly impact patient and/or operator safety.

Table 1: Packaging and Safety Symbols

Safety Notice	Meaning
 DANGER	A red DANGER symbol is used to identify conditions or actions for which a specific hazard is known to exist. These conditions or actions will cause severe personal injury, death or substantial property damage if the instructions are not followed.
 WARNING	An orange WARNING symbol is used to identify conditions or actions for which a specific hazard is known to exist. These conditions or actions may cause severe personal injury or substantial property damage if the instructions are not followed.
 CAUTION	A yellow CAUTION symbol is used to identify conditions or actions for which a potential hazard may exist. These conditions or actions may cause minor personal injury or property damage if the instructions are not followed.
	IEC 60601 Type BF (body field) Applied Part.
	Class II (double-insulated) electrical equipment.
	Electrostatic Discharge (ESD) sensitivity.
	Consult operating instructions before using.
	Consult maintenance instructions before performing any maintenance on the system.
	Manufacturer of the device.

Safety Notice	Meaning
	NRTL (Nationally Recognized Testing Laboratories) safety testing and compliance.
	Conformity with the essential requirements set out in the European Directives.
	Authorized representative for Europe.
	Range of acceptable temperature conditions for shipping and storage.
	Acceptable range of relative humidity for shipping and storage.
	Acceptable range of atmospheric pressure for shipping and storage.
	Do not cross or loop cables and/or ECG leads; arcing and patient burns can result.
	Ensure cables or body parts are not pinched during patient support travel into or out of the scanner.
	The crossed out wheeled bin symbol is used to mark products that should not be disposed with general household waste, but collected separately for reuse or recycling. No part of the Tabletop system shall be disposed of in land fill. Return the device to Sentinelle Medical for disposal.
	Locked.
	Unlocked.

Safety Notice	Meaning
     	Use this riser on systems with a flat patient table only (GE Discovery 750w System with GEM Option). Do not use this riser on systems with a curved patient table (Discovery MR750w without GEM Option, Discovery MR750, Signa HD, or Signa HDx/HDxt).
  	For 3T scanners with a curved patient table, (Discovery MR750, Discovery MR750w without GEM Option, Signa HD, or Signa HDx/HDxt scanners), place the Patient Support directly on the patient table. For Discovery MR750w with GEM Option scanners, which have a flat patient table, always place the riser between the Patient Support and the patient table. Do not place the Patient Support directly on the patient table.
L	Left.
M	Medial.
R	Right.

Other Manual Conventions

The following styles are used throughout this manual.

- A *NOTE* provides additional helpful information. It may emphasize information regarding special tools or techniques, items to check before proceeding, or factors to consider about a concept or task. Notes are always italicized.
- Information in **Bold Sans-serif** type indicates scanner software text as it appears on the screen.

Patient and System Safety Warnings

The following general warning statements apply to scanning with this system in a magnetic resonance (MR) scanner. For additional safety information, review the warnings and safety standards in your MR system documentation.

Table 2: General Warning Statements

Safety Notice	Meaning
	Patients must not form a loop with any body parts while scanning. For example, they cannot touch any part of the right hand or arm to any part of the left hand or arm. This could cause a burn at the point of contact. Use pads to prevent any part of the patient's body from touching metal on the patient table. Educate the patient accordingly and check the patient's position immediately before the scan.
 WARNING	This coil system is only compatible with GE 3.0T MRI System with P Ports Scanners. Remove unused or improperly connected coils, cables, and gating leads from the MR scanner before performing a scan. Cables can become hot if they are left on the patient table when they are not connected to the scanner. Ensure the patient is not wearing damp clothing. This may interfere with proper operation of the system. All clothing containing metallic objects must be removed from the patient prior to exam. The use of medicinal products in transdermal patches may cause burns to underlying skin. All transdermal patches must be removed and the skin thoroughly cleaned prior to using the system as per your institution's MRI safety protocol. Check that the cable or lead(s) do not touch the sides of the MR scanner. Exposure of the patient to radio frequency magnetic fields may increase the potential for local excessive heating of the patient. Immediately stop the MR scan acquisition if the patient complains of a tingling and/or heating sensation. All personnel using the Tabletop must receive instruction in the proper connection and handling of the system and should be familiar with this User Manual.

Safety Notice	Meaning
 WARNING	Auxiliary equipment (such as gating equipment, vital sign monitoring systems and RF coils) that has not been specifically tested and approved for use in an MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of this system.
	Coils are not suitable for use in the presence of flammable anaesthetics in combination with air, oxygen or nitrous oxide.
	Service personnel must have specialized training to ensure the safe operating condition of the coil and table system. Only properly trained and qualified personnel are authorized to service any components.
	The patient's body and extremities must not contact the inner surface of the MR bore. Burning may occur.
	Patients that are sedated, unconscious or are experiencing a loss of feeling in any body part should not be scanned. These patients would not be able to alert the operator as to excessive heating and associated tissue damage.
	Before using the Tabletop, check to ensure there is no external damage. Do not use a coil if the housing or cable is broken. Do not use coils, cables, or gating leads with exposed metal surfaces or abraded insulation.
	Modifications made to the system that are not authorized by Sentinelle Medical may void the warranty. Changes to the system may impact the safety of the system.
 CAUTION	Ensure the Tabletop cable is located centrally in the scanner. Do not run the Tabletop cable along the edge of the bore.
	Connect the coil cables to the connector only as prescribed in this manual.
	Position the patient and the coils and cable as prescribed in this User Manual.
	Ensure that the coil cable does not directly contact any part of the patient.
	For Discovery MR750w with GEM Option scanners, always place the riser between the Patient Support and the patient table. Do not place the Patient Support directly on the patient table.

Safety Notice	Meaning
 CAUTION	Ensure the Tabletop Cable is not connected to the LPCA when lowering the scanner table.
	The Tabletop cable must be covered with the body pad before operation.
	To avoid tipping, use the Velcro straps to secure the Tabletop.
	Use only MR-compatible ECG leads and electrodes, such as those provided or specified by the equipment manufacturer. Never use electrodes that have passed their expiration date.
	Keep limbs away from the gap between the Tabletop and the scanner while raising and lowering the MR patient table.
	Do not reach under any of the guards while the MR patient table is being raised and lowered.
	Ensure the patient's feet do not overhang the end of the patient table while docking to the scanner.
	To avoid possible patient injury, do not use the automatic button to advance the patient into the scanner. Use the manual advance button and watch for potential contact of the patient with the bore as they advance into the scanner.
	Avoid using the Tabletop system adjacent to or stacked with other equipment. If this is not avoidable, observe the equipment to verify that it operates normally.
	Do not cross or loop cables and/or ECG leads; this may result in arcing and patient burns.

Contacting Sentinelle Medical

Use the following information to contact Sentinelle Medical to report an incident or for technical support:

Phone (Including Technical Support): 1-866-735-3744 (North America)
1-416-366-4994 (International)

Fax: 1-647-258-0299

Email: service@hologic.com

Service hours are 8:00am - 5:00pm Eastern Standard Time (EST), Monday-Friday.

Reporting Incidents

Contact Sentinelle Medical immediately to report any incident and/or injury to a patient, operator or maintenance employee that occurred as a result of using the Tabletop.

If an accident occurs as a result of using the Tabletop, do not operate the equipment until an authorized investigation has been conducted.

For contact information, see Contacting Sentinelle Medical above.

Electrostatic Discharge (ESD) Safety



Do not touch the pins of connectors identified with the ESD warning symbol.

Do not use these connectors without taking ESD precautions. This includes:

- methods to prevent electrostatic charge build-up (for example, air conditioning, humidification, conductive floor coverings, non-synthetic clothing),
- discharging your body to earth or a large metal object,
- grounding yourself by means of a wrist strap to earth.

Electromagnetic Environment Information

This equipment receives RF electromagnetic energy. The system operates at a frequency of 127.73 MHz, over a bandwidth of ± 360 kHz. Other equipment may interfere with the Tabletop even if it complies with CISPR emission requirements.

The Tabletop must be used in the electromagnetic environment described below.

Table 3: Electromagnetic Environment (Emissions)

Emissions test	Classification	Electromagnetic Environment Guidance
RF emissions CISPR 11	Group 2	The Tabletop must receive electromagnetic energy to perform its intended function. Nearby electronic equipment may be affected.
RF emissions CISPR 11	Class A	The Tabletop must be used only in a shielded location with a minimum RF shielding effectiveness and, for each cable that exits the shielded location, a minimum RF filter attenuation of 20 dB over a minimum frequency range of 30 MHz to 300 MHz.
Harmonic emissions IEC 61000-3-2	Not Applicable	
Voltage fluctuations/flicker emissions IEC 61000-3-3	Not Applicable	
NOTE: It is essential that the actual RF shielding effectiveness and filter attenuation of the shielded location be verified to ensure that they meet or exceed the specified minimum values.		

Table 4: Electromagnetic Environment (Immunity)

Immunity test	IEC 60601 test level	Classification level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	±6 kV contact ±8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	Not Applicable	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV line(s) to line(s) ±2 kV line(s) to earth	Not Applicable	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11 UT = 230 Vac	<5% UT (>95% dip in UT) for 0,5 cycle 40% UT (60% dip in UT) for 5 cycles 70% UT (30% dip in UT) for 25 cycles <5% UT (>95% dip in UT) for 5 sec	Not Applicable	Mains power quality should be that of a typical commercial or hospital environment. If the Tabletop must operate continuously during power mains interruptions, it must be powered from an uninterruptible power supply or a battery.
Power frequency (50 Hz) magnetic field IEC 61000-4-8	3 A/m	Not Applicable	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

Table 5: Electromagnetic Environment (Conducted/Radiated)

Immunity test	IEC 60601 test level	Classification level	Electromagnetic Environment Guidance
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 Vrms	The Tabletop must be used only in a shielded location with a minimum RF shielding effectiveness and, for each cable that enters the shielded location, a minimum RF filter attenuation of 20 dB from a minimum frequency range of 30 MHz to 300 MHz (See the previous table.) Field strengths outside the shielded location from fixed RF transmitters, as determined by an electromagnetic site survey, should be less than 3 V/m.
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2,5 GHz	3 V/m	Interference may occur in the vicinity of known RF transmitting devices and equipment marked with the following symbol: 

NOTE: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. An electromagnetic site survey will assess the electromagnetic environment due to fixed RF transmitters. If the measured field strength in the location where equipment is used exceeds the applicable RF compliance levels, the equipment should be observed to verify normal operation. If abnormal performance is observed, you may need to re-orient or relocate the equipment or use a location with higher RF shielding effectiveness and filter attenuation.

Chapter 1 The Tabletop System

General System Function and Components

The Tabletop is a dedicated breast MR imaging coil and interventional system for 3.0T GE MRI Systems with P Ports. It consists of a patient support with breast immobilization system and an attached 2-, 10, or 16 channel coil system.

For clinical imaging, coil housings are placed next to the tissue to help minimize motion artifacts due to patient motion during scanning.

For interventional procedures, sterile, single-use disposable biopsy grids* are used to both immobilize the breast and provide access windows for biopsy needle guidance.

System Classification



The Tabletop System is classified according to IEC 60601-1 as follows:

- Type BF Applied Part
 - Degree of Protection against Ingress of Water: IPX0
- 
- Equipment not suitable for use in the presence of a Flammable Anaesthetic mixture with Air or Oxygen or Nitrous Oxide
 - Mode of Operation: Continuous operation

Indications and Contraindications for Use

Indications for Use

The Tabletop is designed to provide magnetic resonance (MR) images of breast anatomy when used in conjunction with an MR scanner. The images can be interpreted by a trained physician. When used with a single-use sterile biopsy grid*, the Tabletop permits access to the breast anatomy for biopsy and localization procedures that can be performed by a trained physician.

* NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

Contraindications for Use

The following contraindications for use relate to the strong magnetic field of the MR scanner:

Table 6: Contraindication Safety Notices

 DANGER	Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants such as cardiac pacemakers, or who rely on electrically, magnetically or mechanically activated external life support systems. The magnetic and electromagnetic fields produced by the MR scanner and coil may interfere with the operations of these devices. Scanning patients with intracranial aneurysm clips is contraindicated unless the physician is certain that the clip is not magnetically active.
 CAUTION	Examination by MR equipment requires particular caution in the following cases: • patients with implanted surgical clips (haemostatic clips) or other ferromagnetic materials, which the magnetic field may dislodge • patients with certain types of intrauterine devices (IUDs) • patients engaged in occupations or activities which may cause accidental implantation of ferromagnetic materials, or who may have embedded metal fragments from military activities • patients with permanent (tattoo) eye-liner or with facial make-up (severe eyelid irritation has been reported) • patients with compromised thermoregulatory systems • patients with metal implants (these may cause artifacts in the diagnostic images) • patients with implanted prosthetic heart valves • patients who are pregnant

See also Safety Notices for Interventional Procedures on page 55.

Specifications

Coil System Specifications

- Field Strength: 3.0T
- Frequency: 127.73 MHz

Patient Support Specifications

- For use with 3.0T GE MRI Systems with P Ports
- Maximum patient weight: 158.8kg (350lb)
- Maximum load on each patient support arch: 45.4kg (100lb)

For details on correct patient positioning to support the maximum patient weights listed, please refer to the Patient Positioning and Preparation Guidelines Chapter, beginning on page 45.

Compatible Biopsy Devices

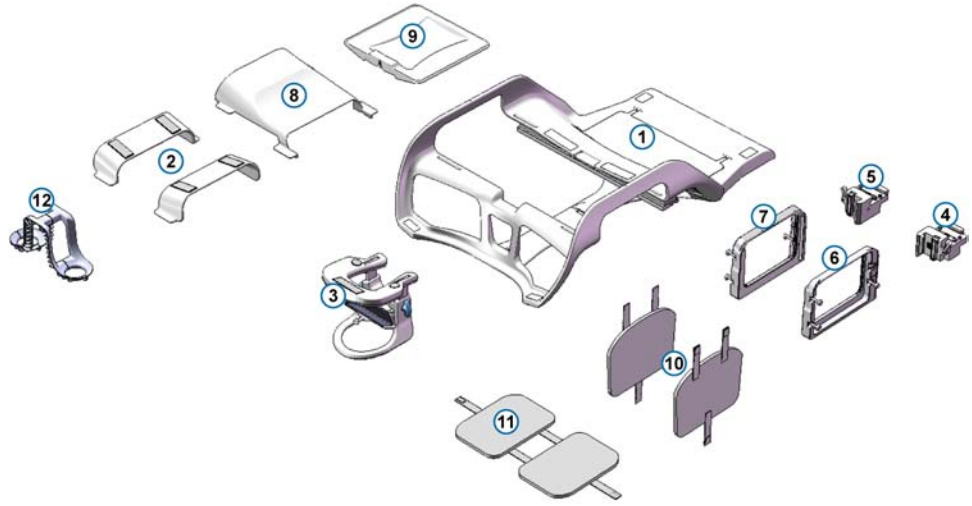
The following MRI-compatible biopsy devices are mechanically compatible with the Tabletop.

- Suros ATEC
- Suros ATEC Petite
- SenoRx EnCor 7G
- SenoRx EnCor 10G
- SenoRx EnCor 14G
- Mammotome J&J 8G 115mm and 145mm
- Mammotome J&J 11G 115mm and 145mm
- Bard VACORA 10G 118mm and 140mm
- Bard VACORA 14G 116mm and 138mm
- WireLoc 14G and 18G*

** Any gauge that fits in the needle block is compatible, however only 14G and 18G are options in the Sentinelle Medical Aegis software product.*

System Components

The following figures show the Vanguard components in each type of system.

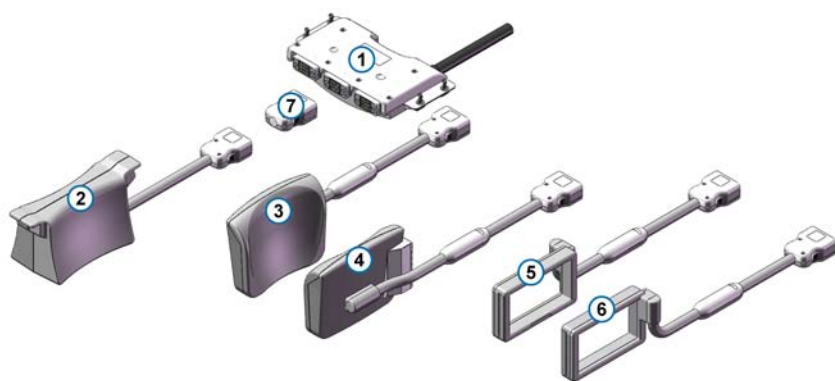


#	Component	Qty	Complete System	16-Ch Imaging-only	10-Ch Option
1	Patient Support	1	✓	✓	✓
2	Shoulder Support with Velcro kit	2	✓	✓	✓
3	Headrest	1	✓	✓	✓
4	Left Horizontal Slider	1	✓	✓	✓
5	Right Horizontal Slider	1	✓	✓	✓
6	Left Immobilization Frame	1	✓	✓	✓
7	Right Immobilization Frame	1	✓	✓	✓
8	Contralateral Breast Support	1	✓		✓
9	Catchment Tray	1	✓		✓
10	Lateral Asset Pad	2	✓	✓	
11	Medial Asset Pad	1	✓	✓	
12	Phantom Positioner	1	✓	✓	✓

#	Component	Qty	Complete System	16-Ch Imaging-only	10-Ch Option
13	Immobilization Plate	2	✓		✓
14	Anti-tip Velcro Straps (not shown)	2*	✓	✓	✓

Figure 1: Patient Support and Accessory Components

* The anti-tip Velcro straps come in a set of five; use two to secure the Tabletop to the patient table.



#	Component	Qty	Complete System	16-Ch Imaging-only	10-Ch Option
1	Interface Box	1	✓	✓	✓
2	Medial Array Coil	1	✓	✓	✓
3	Left Lateral Array Coil	1	✓	✓	
4	Right Lateral Array Coil	1	✓	✓	
5	Left Single Loop Coil	1	✓		✓
6	Right Single Loop Coil	1	✓		✓
7	Medial Plug	1	✓		✓

Figure 2: RF Components

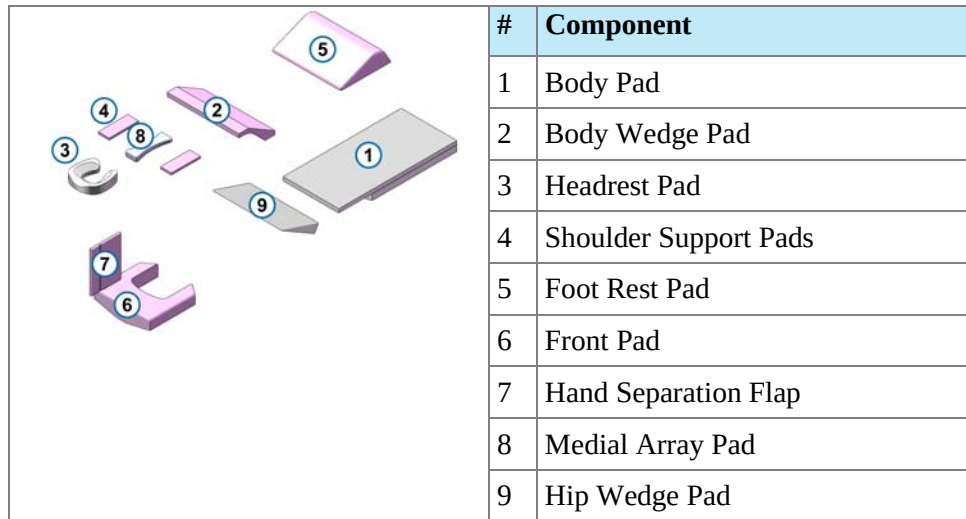


Figure 3: Padding Components

NOTE: For instructions on how to arrange the padding for optimal patient comfort, see the Patient Positioning and Preparation Guidelines Chapter, beginning on page 45.

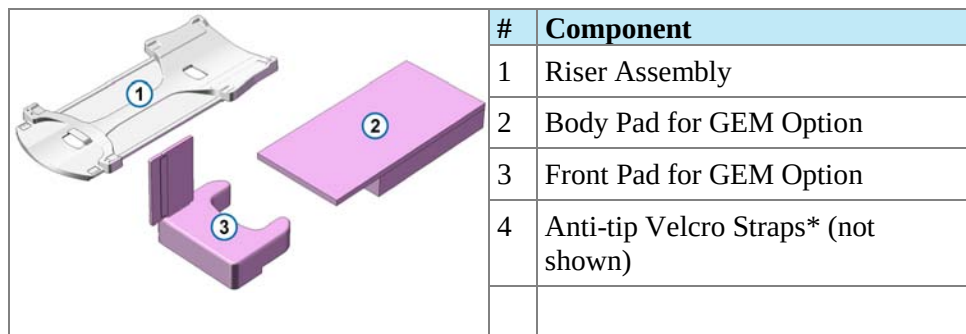


Figure 4: Riser Components for the GE Discovery 750w System with GEM Option

** The anti-tip Velcro straps come in a set of five; use four to secure the Tabletop to the GE Discovery 750w System with GEM Option patient table.*

Patient Support Setup

The patient support setup procedures are different for scanners with a curved patient table (GE Discovery 750 System and GE Discovery 750w System without GEM Option) and scanners with a flat patient table (GE Discovery 750w System with GEM Option). Be sure to follow the steps that apply to your scanner.

Patient Support Setup for the GE Discovery 750 System and GE Discovery 750w System without GEM Option

Place the Tabletop on the 3.0T GE MRI System patient table with the foot end toward the bore so the Tabletop cable, located at the foot end of the patient support, can be securely plugged into the MR scanner.



Figure 5: Patient Support on GE Patient Table



Do not use the riser on the GE Discovery 750 System and GE Discovery 750w System without GEM Option. Using the riser on these systems may damage the equipment and potentially injure patients.

Using the Anti-Tip Velcro straps

To prevent the patient support from tipping, ensure the anti-tip Velcro straps are properly attached before allowing a patient to mount the patient support. Insert a Velcro strap into the slot on each side of the 3.0T GE MRI System table, then slide and fasten them to the Tabletop.



Figure 6: Securing the Patient Support with the Anti-Tip Velcro straps

Patient Support Setup for the GE Discovery 750w System with GEM Option



You must use the riser on scanners with a flat patient table (GE Discovery 750w System with GEM Option). Failure to use the riser may cause the Tabletop to tip over, potentially injuring patients.

Do not use the riser on scanners with a curved patient table (GE Discovery 750 System and GE Discovery 750w System without GEM Option).

To set up the Tabletop on the GE Discovery 750w System with GEM Option:

1. Place the riser on the patient table with the rounded end of the riser pointing away from the bore.



Figure 7: Riser on the Patient Table

2. Place the patient support on the riser with the head end of the patient support toward the rounded end of the riser.
3. Attach the patient support to the riser using the Velcro straps (circled in the figure below, left).
4. Secure the riser to the patient table by fastening the anti-tip Velcro straps to the Velcro pads at the head end of the riser. To provide additional stability, you can also fasten anti-tip Velcro straps to the Velcro on the reverse side of the straps that connect the patient support to the riser.

For more information about using the anti-tip Velcro straps, see Using the Anti-Tip Velcro Straps above.

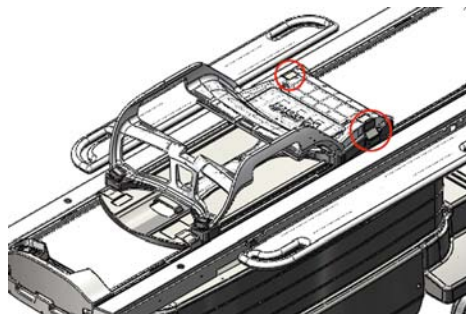


Figure 8: Patient Support Connected to the Riser (Velcro Straps Circled)

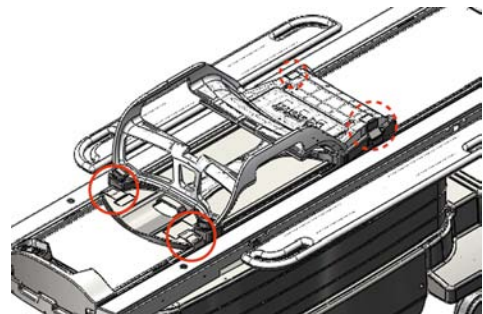


Figure 9: Velcro straps to Connect the Riser and Patient Support to the Patient Table (Velcro Straps Circled)

NOTE: If you will be using the Tabletop on a GE Discovery 750w System with GEM Option only, you can leave the patient support connected to the riser and store them together.

Catchment Tray

For unilateral and bilateral interventional procedures the catchment tray must be in place. This is a safety mechanism ensuring the MR guided needle device can safely be directed into the breast tissue. During interventional procedures, you may want to drape the catchment tray with an absorbent pad to catch any spills that happen during the procedure.



Figure 10: Inserting the Catchment Tray

Using the Headrest

The headrest cushion is attached to a movable stand. To adjust the height, unlock the headrest by turning the knob on the side, as shown in the figure below on the left. Adjust the angle of the headrest by pulling out the handle at the front as shown in the figure below on the right.

A mirror is available to allow the patient to see outside the bore during scanning. Adjust the mirror position after the height and angle of the headrest have been adjusted for comfort.

If the headrest is in its lowest height and angle position, remove the mirror by pulling it directly away from the headrest to avoid interference.



Figure 11: Adjusting Headrest Height



Figure 12: Adjusting Headrest Angle

The Coils

The Tabletop coils consist of a 16-channel configuration, and an interchangeable 10-channel and 2-channel configuration for interventional work. A 10-channel configuration is also available for interventional work that consists of an 8-channel medial array with two single loop coils. The coils can be changed or reconfigured, depending upon the particular imaging requirement.

An Imaging-only option comprises a 16-channel coil set only.

NOTE: To upgrade to a complete system, contact your Sentinelle Medical sales representative.

Coil Naming

For consistency and ease of use, this manual uses simplified coil names, not the coil names as they are listed on the coil labels. The following table maps the technical name of each coil as it appears on the coil label to the common name for the coil used in this manual.

Table 7: Coil Name Cross-reference

Coil Label Name	Coil Common Name
Lateral Array Coil Right	Right Lateral Array Coil
Lateral Array Coil Left	Left Lateral Array Coil
Medial Array Coil	Medial Array Coil
Lateral Single Loop Coil Left	Left Single Loop Coil
Lateral Single Loop Coil Right	Right Single Loop Coil

Coil Configurations

The scanner will automatically select the appropriate configuration when a valid coil combination is connected. The following table describes the various coil arrangements, available for specific procedural requirements.

Table 8: Available Coil Configurations

Procedure	Receptacle			Coil Configuration on MRI	Complete System	16-Ch Imaging-only	10-Ch system
	L	M	R				
16-Ch Bilateral Imaging	Left Lateral Array Coil	Medial Array Coil	Right Lateral Array Coil	Select coil 16Ch Breast Array by Sentinelle Use: Breast SMI	✓	✓	

Procedure	Receptacle			Coil Configuration on MRI	Complete System	16-Ch Imaging-only	10-Ch system
	L	M	R				
16-Ch Unilateral Imaging	Left Lateral Array Coil	Medial Array Coil	Right Lateral Array Coil	Select coil 16Ch Breast Array by Sentinelle Left Breast use: LtBreast SMI Right Breast use: RtBreast SMI	✓	✓	
10-Ch Bilateral Imaging/ Intervention	Left Single Loop Coil	Medial Array Coil	Right Single Loop Coil	Select coil 10ch Biopsy Breast Array by Sentinelle Use: 10Brst Int SMI	✓		✓
10-Ch Unilateral Imaging	Left Single Loop Coil	Medial Array Coil	Right Single Loop Coil	Select coil: 10ch Biopsy Breast Array by Sentinelle Left Breast use: 10BrstLt Int SMI Right Breast use: 10BrstRt Int SMI	✓		✓
Unilateral Intervention	Left Single Loop Coil	Medial Plug	Right Single Loop Coil	Select coil 2ch Biopsy Breast Array by Sentinelle Use: 2Brst Int SMI	✓		✓

Coil Setup

The following figures show the Tabletop's general set up for bilateral breast imaging, unilateral interventional and bilateral interventional procedures.

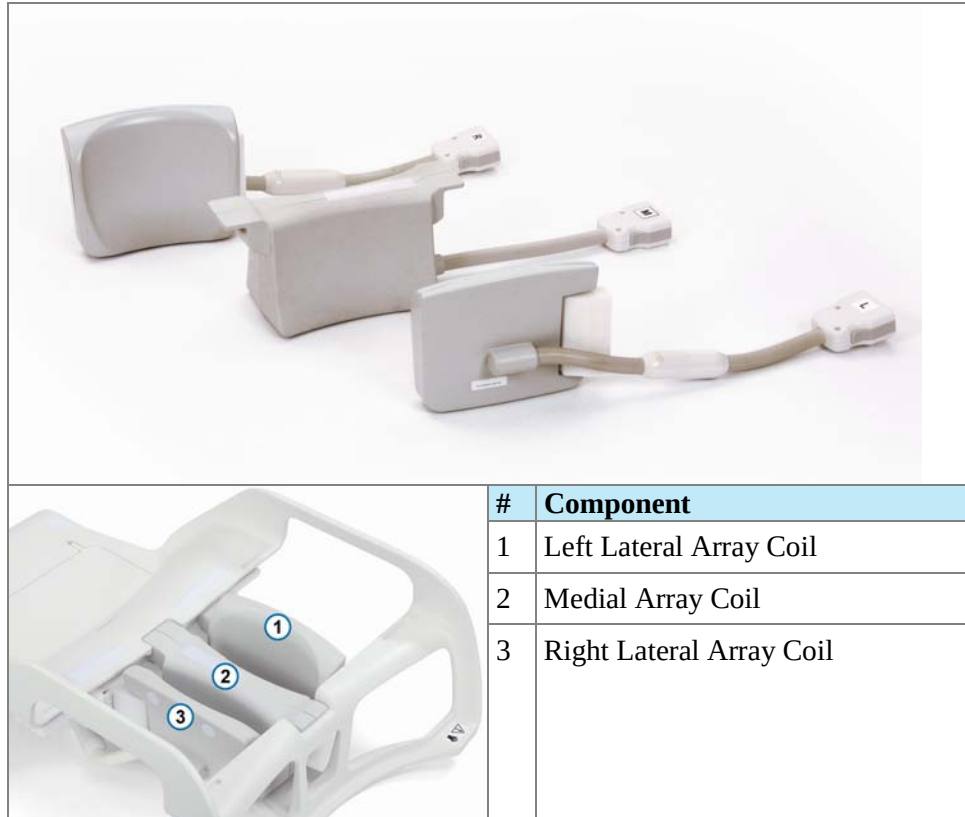


Figure 13: Bilateral Imaging Components and Setup (Without Padding; Asset Pads Not Shown)

Using the Asset Pads

For bilateral imaging procedures, attach the asset pads to the medial and lateral coils to ensure optimal imaging performance.

To attach the lateral asset pads to the lateral array coils, connect the Velcro tabs on the asset pads to the corresponding Velcro pads on the back side of the coils.



Figure 14: Asset Pad Attached to a Lateral Array Coil (Front and Back View)

To attach the medial asset pad to the medial array coil, wrap the medial asset pad around the medial array coil and connect the Velcro tabs to each other beneath the coil.



Figure 15: Asset Pad Attached to the Medial Array Coil

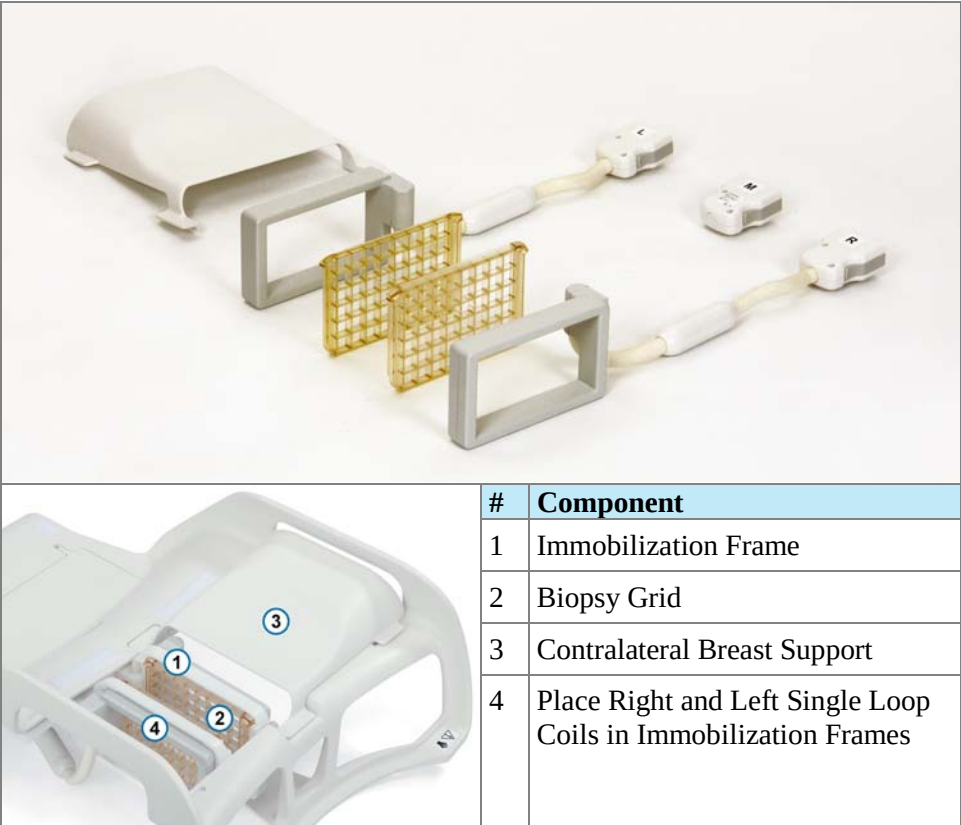


Figure 16: Unilateral Intervention Setup and Components (Without Padding)

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

About the Contralateral Breast Support

The contralateral breast support holds the contralateral breast away to improve medial access during unilateral interventional procedures. The design of the breast support allows the breast of interest to fall further into the aperture for increased lateral posterior access to the breast during intervention.

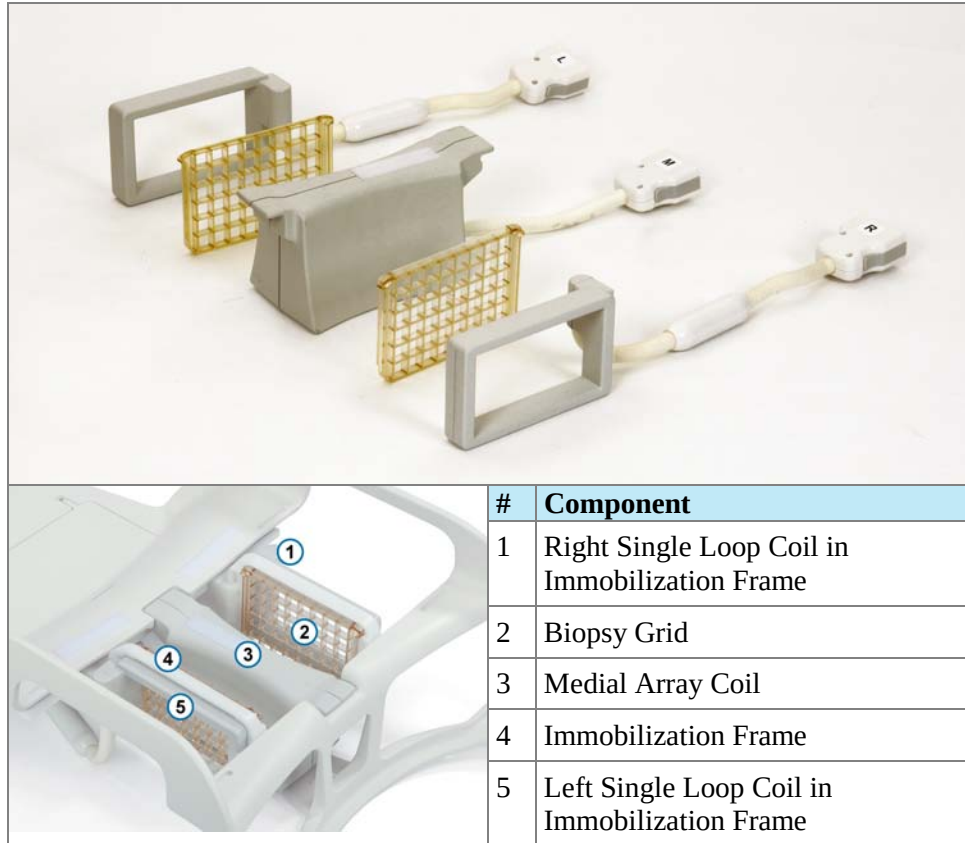


Figure 17: Bilateral Intervention Setup and Components (Without Padding)

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

The Vanguard Cable

The Tabletop cable connects the Tabletop to the scanner.



A "System Ready" green light appears on the Interface Box when the Tabletop cable is properly plugged in. Do not scan if the cable is unplugged.

The Tabletop cable must be covered with the body pad before operation.



Figure 18: Covering the Tabletop Cable with the Body Pad

Inserting and Connecting the Coils

To insert the lateral array coils, insert each coil into the compression slider by rotating the coil housing into place. Continue to push medially until the hard stop extends past the coil. Insert the medial coil array in the center of the patient support, between the left and right lateral coils.

To insert the single loop coils for interventional procedures, first insert an immobilization frame into one or both sliders, as appropriate for the procedure, then place the single loop coils in the frames.

Plug the coils into the interface box. The plugs are keyed so that the coils can only be connected in the correct orientation.



Figure 19: Inserting a Lateral Array Coil



Figure 20: Connecting the Lateral Array Coil



Figure 21: Inserting the Medial Array Coil

Using the Immobilization System

The Vanguard immobilization system holds the breast immobile during scanning. The following figure illustrates the Vanguard immobilization system and components.

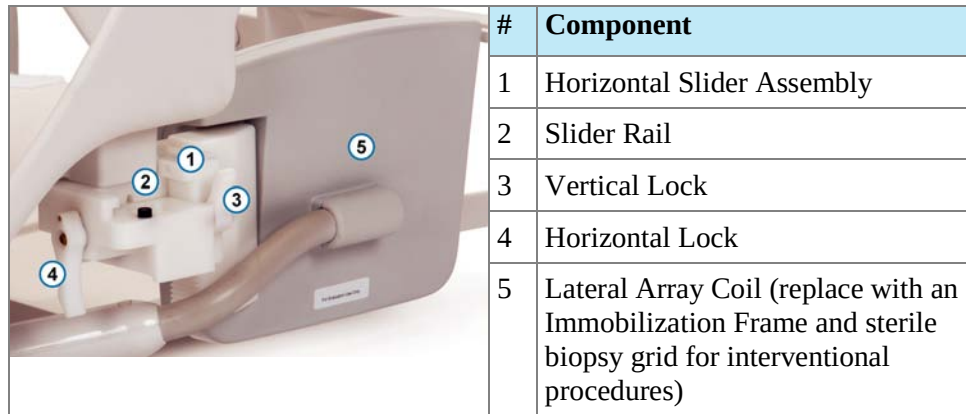


Figure 22: The Tabletop Immobilization System

The lateral array coils and immobilization frames can be adjusted both vertically and horizontally (anterior/posterior and medial/lateral). Each direction has its own locking mechanism (See the figures below).

To slide a coil or immobilization frame up or down, release the vertical lock and move the coil or frame up or down in its slider. When the coil or frame is in the desired vertical position, push the lever to lock it in place.

To slide a coil or immobilization frame left and right, release the horizontal lock and move the slider left or right along the slider rail. Push the lever to lock the coil or frame when it is in the desired position.



Figure 23: Adjusting Vertical Lock

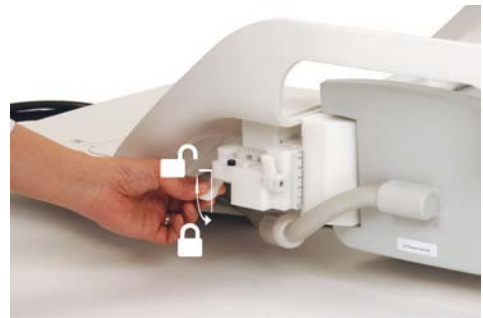


Figure 24: Adjusting Horizontal Lock

The coils or frames can be removed from the slider by pressing the actuator and rotating the coil or frame laterally to release it from the slider.



Figure 25: Releasing an Immobilization Frame

Coil Service

The coils cannot be serviced on site, they must be returned for servicing by a trained representative. Contact your Sentinelle Medical service representative if your coils require servicing (see below).

If your system has been ordered from Sentinelle Medical, use the following contact information:

Phone: (North America) 1-866-735-3744 or (International) 1-416-366-4994

Email: service@hologic.com

Address: Sentinelle Medical, a Division of Hologic
Suite 800, 555 Richmond Street West
P.O. Box 301 Toronto, Ontario, Canada
M5V 3B1

If your system has been ordered from GE Healthcare, contact your GE representative.

NOTE: Biopsy grids are not supported by GE. For any biopsy grid inquiries, contact Sentinelle Medical.

Coil Storage

For coil storage, the Sentinelle Medical Companion Cart (part number 4000474-51) is available for purchase. The cart can be used to store coils and coil

components. The detachable cabinet can be used as a mobile tray during interventional procedures. Please contact your Sentinelle Medical sales representative for purchasing details.



Figure 26: Sentinelle Medical Companion Cart

Chapter 2 Quality Assurance (QA)

Instructions for MR Imaging on Phantoms

Sentinel Medical recommends that the coils be checked for imaging integrity on a regular basis, by performing the Quality Assurance (QA) procedure. The phantoms used for the QA scan are part of GE's System Phantom Set (GE Model 46-317586G1). The instructions below apply only to the Sentinel Vanguard system.

NOTE: If small bubbles are apparent on the sides of the phantom, tap the outside of the phantom until these bubbles disappear.

NOTE: Do not use the asset pads when performing the QA scan.

Test 1: 16-Channel Bilateral Breast SNR Verification

To run the 16-channel bilateral breast SNR verification:

1. Place the Tabletop on the MR scanner table.
2. Install and connect the 16-channel coil set and plug the Vanguard cable into the receptacle.
3. Move the coils to their most lateral position and raise them to their highest vertical position.
4. Place the phantom positioner on the medial coil as shown in the figure below.



Figure 27: Phantom Positioner Setup for 16-Channel Coils

5. Add the spherical GE phantoms and move the coils to their lowest position. Adjust the coils medially so that they lightly touch the phantoms as shown in the figure below.



Figure 28: Phantom Setup for 16-Channel Coils

6. Landmark using the notch on the edge of the patient support as shown below.

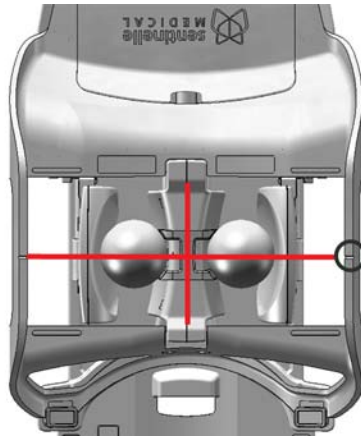


Figure 29: Landmark on the Phantoms

7. Do not start a new patient. Instead, open the **Tools** box and start the **Service Browser**.
8. In the **Service Browser**, select the **Image Quality** tab.
9. In the side menu, select **Multi-Coil QA Tool**, then **Click here to start this tool**. The application starts, as shown below.

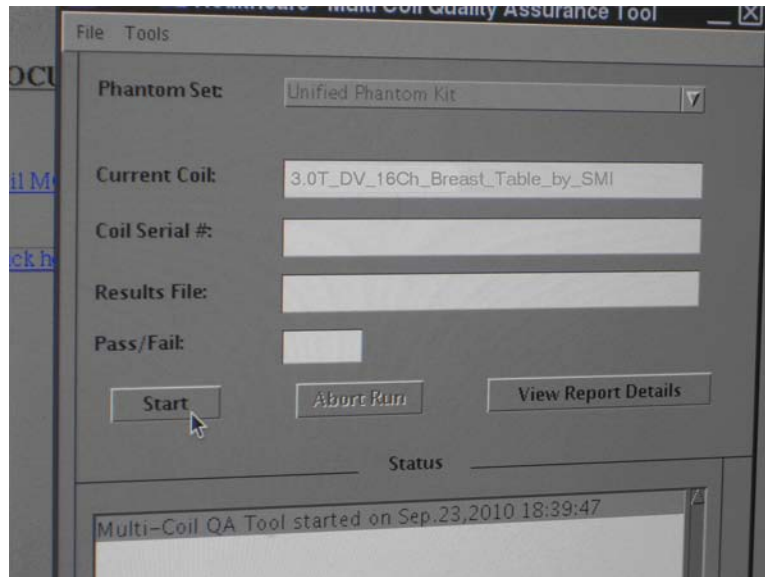


Figure 30: Screen Capture - MCQA Tool Start

10. Click **Start** and answer **Yes** in the dialog that appears.

The MCQA Tool will run for approximately 10 minutes, testing each channel in the system independently. When the tests are complete, the MCQA Tool displays the test results as shown below.

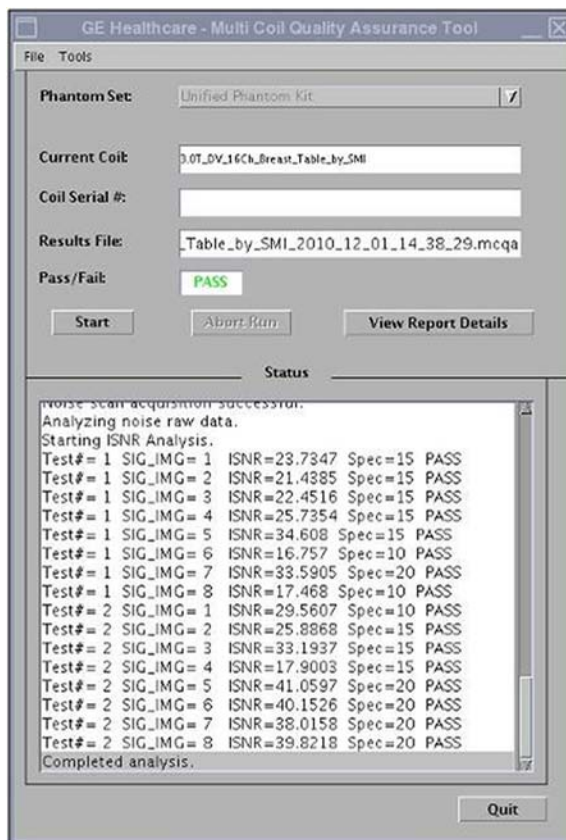


Figure 31: Screen Capture - MCQA Tool Finished

11. If the MCQA indicates that the test failed, check that all coils are connected properly and run the test again. If the test fails a second time, contact your service representative.

Test 2: 10-Channel Bilateral Breast SNR Verification

To run the 10-channel bilateral breast SNR verification:

1. Place the Tabletop on the MR scanner table.
2. Install and connect the 10-channel coil set and plug the Vanguard cable into the receptacle.
3. Move the coils to their most lateral position and raise them to their highest vertical position.
4. Place the phantom positioner on the medial coil as shown in the figure below.



Figure 32: Phantom Positioner Setup for 10-Channel Coils

5. Add the spherical GE phantoms and move the coils to their lowest position. Adjust the coils medially so that they lightly touch the phantoms as shown in the figure below.



Figure 33: Phantom Setup for 10-Channel Coils

6. Landmark using the notch on the edge of the patient support as shown below.

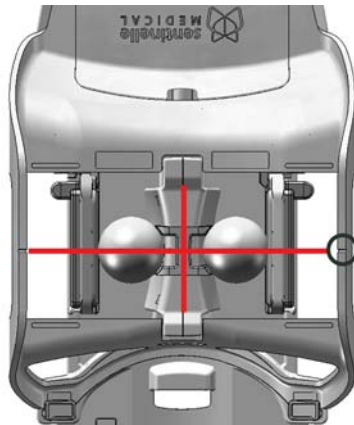


Figure 34: Landmark on the Phantoms

7. Do not start a new patient. Instead, open the **Tools** box and start the **Service Browser**.
8. In the **Service Browser**, select the **Image Quality** tab.
9. In the side menu, select **Multi-Coil QA Tool**, then **Click here to start this tool**. The application starts, as shown below.

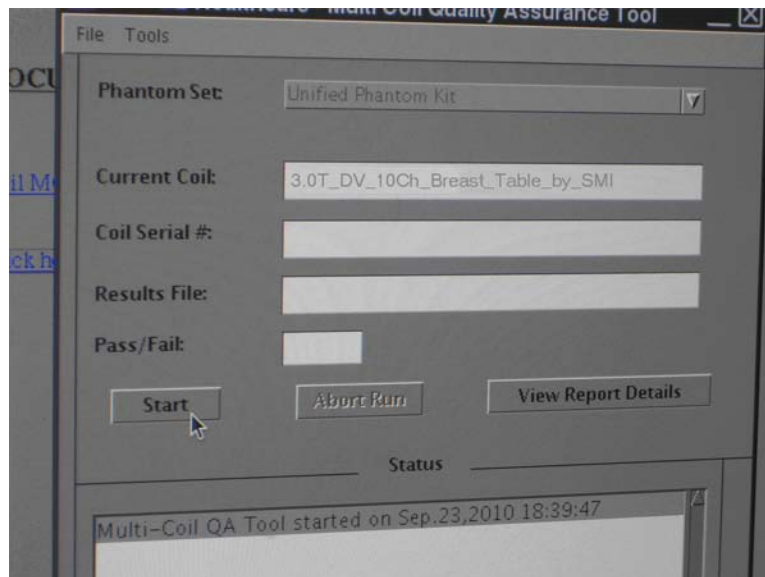


Figure 35: Screen Capture - MCQA Tool Start

10. Click **Start** and answer **Yes** in the dialog that appears.

The MCQA Tool will run for approximately 10 minutes, testing each channel in the system independently. When the tests are complete, the MCQA Tool displays the test results as shown below.

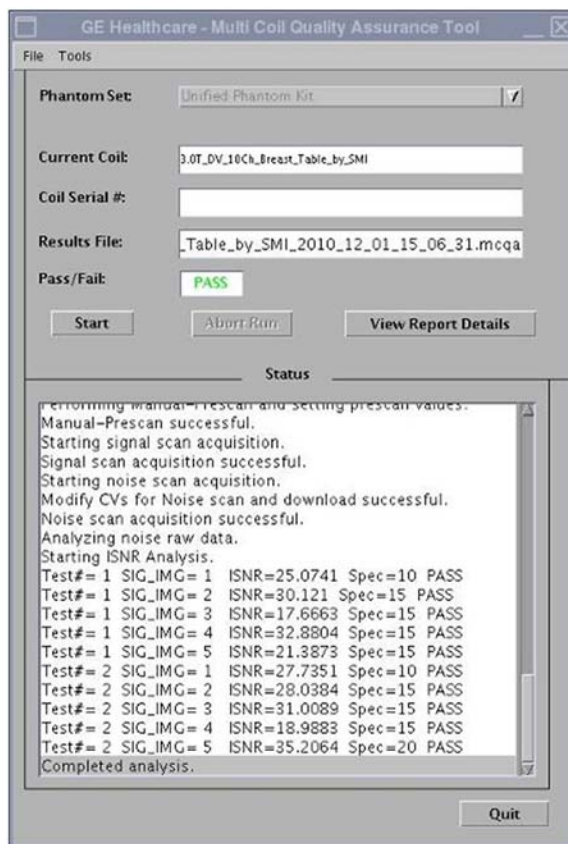


Figure 36: Screen Capture - MCQA Tool Finish

11. If the MCQA indicates that the test failed, check that all coils are connected properly and run the test again. If the test fails a second time, contact your service representative.

Troubleshooting Tips for QA Testing on Spherical Phantoms

Keep the following tips in mind when running the MCQA test:

- Ensure that no patient scan has been started before beginning the MCQA procedure. Exit out of any open patients.
- Landmark the coils before starting the MCQA tool.
- Use only the spherical GE phantoms (GE Model 46-317586G1) and the Sentinelle phantom positioners.
- Verify that the lateral coil A/P position is correct.
- Ensure that all coils are touching the phantoms.

- When the MCQA tool starts, you may see the message shown below. If so, click **Yes** to continue.



Figure 37: MCQA Tool Warning Message

Chapter 3 Patient Positioning and Preparation Guidelines

This chapter describes how to position the Tabletop pads for optimum comfort. See the following figure for the recommended pad placement. In all cases, use the body pad and the body wedge pad. Ensure that the body wedge pad covers the edge of the patient support below the ribs.

NOTE: For the GE Discovery 750w System with GEM Option, always use the thicker riser body pad and front pad.

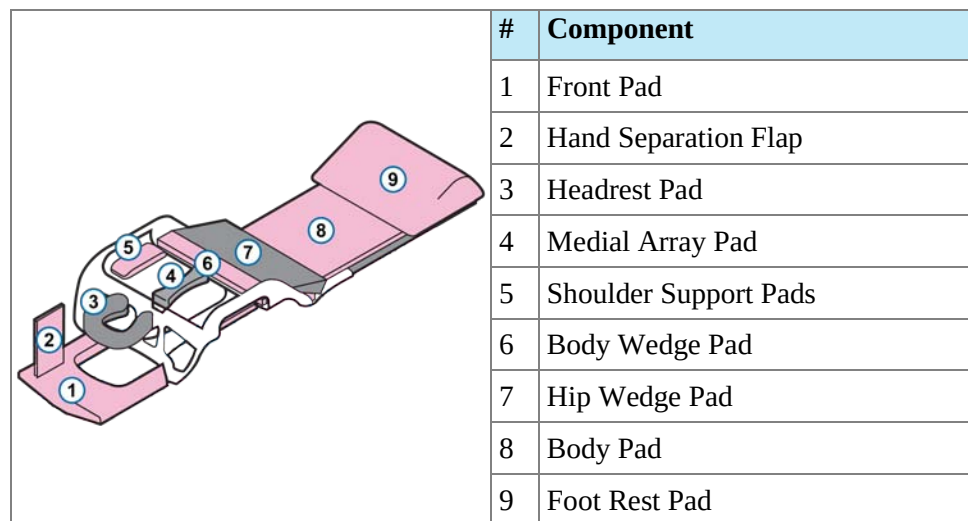


Figure 38: Padding Setup



Figure 39: Patient Lying Prone on the Tabletop

Weight Support

Refer to the Specifications table on page 18 of this manual for maximum patient weight specifications.

To correctly support the patient's weight, the patient support must be properly seated on the MRI table stretcher.



CAUTION

Advise the patient not to place their hands on the coil elements when mounting and dismounting the system. Putting weight on the coils may damage them.

Breast Positioning Guidelines

The figure below depicts guidelines for proper breast positioning. The general rule implies that the inferior border of the breast aligns just within the border of the medial coil. The inferior border of the breast should never be positioned further inferior than the border of the medial coil.

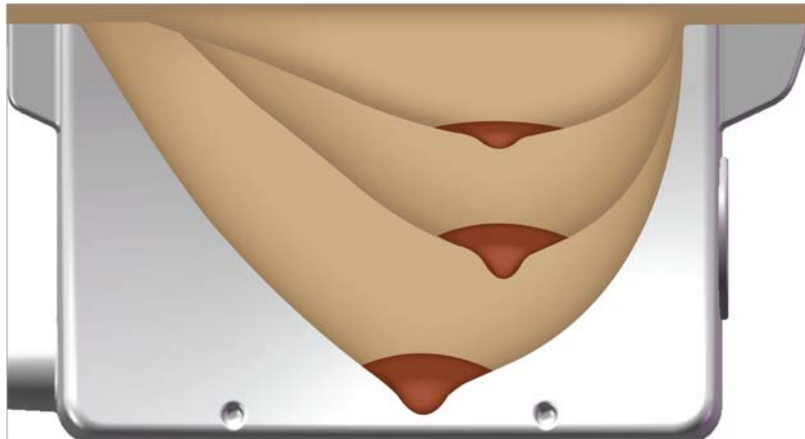


Figure 40: Breast Positioning Guidelines for Small, Medium and Large Breasts

NOTE: It may be helpful to remove the Coil or Immobilization Frame to get a clearer view of how the breast is positioned. Replace the Coil or Immobilization Frame when the breast is positioned correctly.

The figure below depicts the anatomy of the breast. Position the patient such that the scan will include the anatomy of interest.

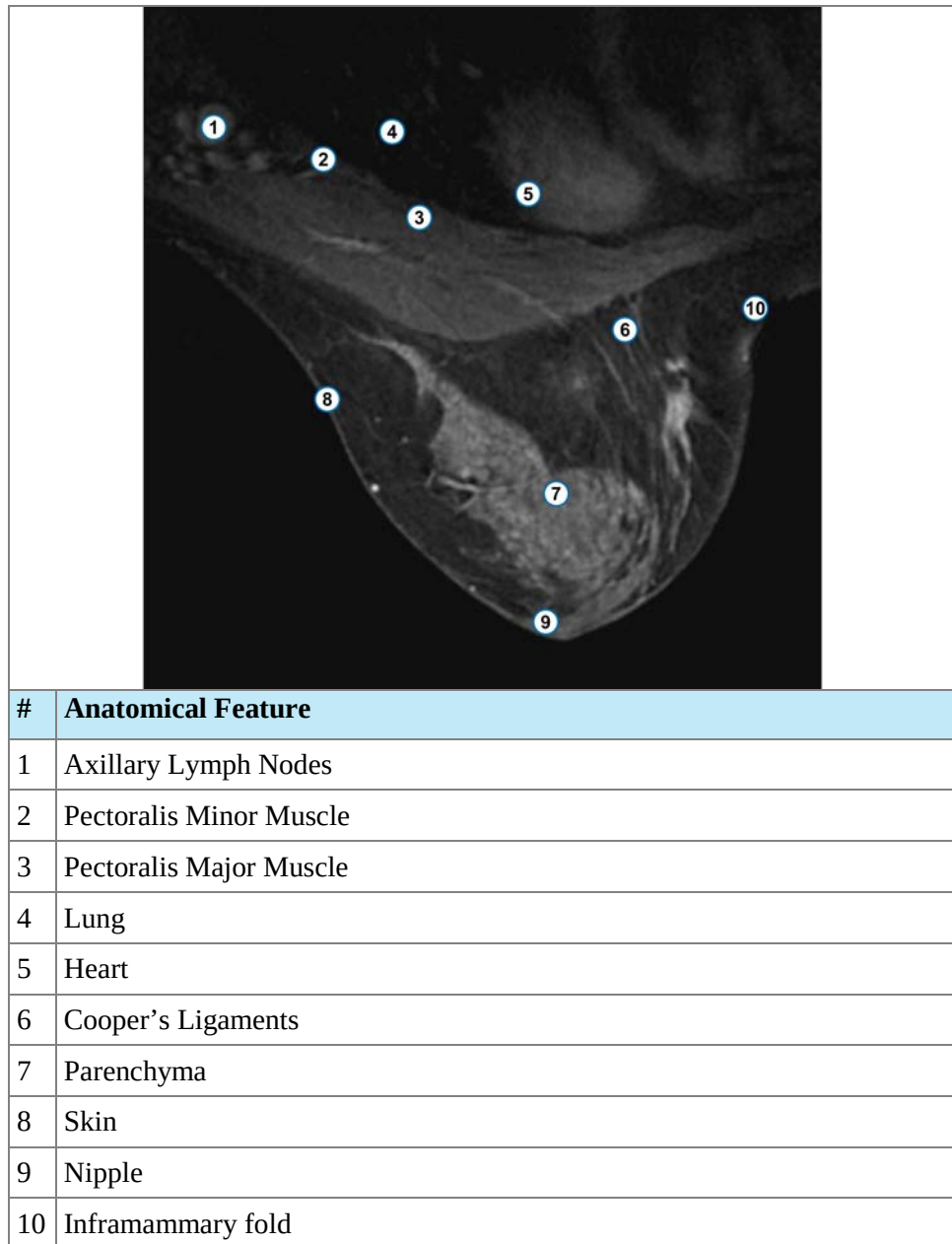


Figure 41: Breast Anatomy

Arm Positioning

Sentinel Medical recommends arms-back positioning. If the patient's arms will not fit between the arches of the patient support, use the arms-forward positioning. Shoulder supports and shoulder support pads provide additional patient comfort by reducing sternum and rib pressure when using the arms-back position, but are not recommended for arms-forward positioning.



If the patient is in the arms-forward position, the hand separation pad must be used between patient's hands so that a loop is not created.



Figure 42: Arms-Back Position



Figure 43: Arms-Forward Position



Figure 44: Incorrect Hand Positioning

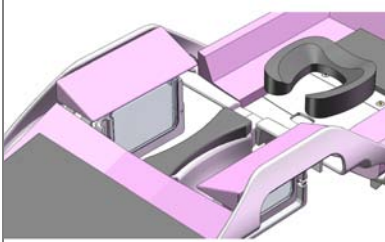
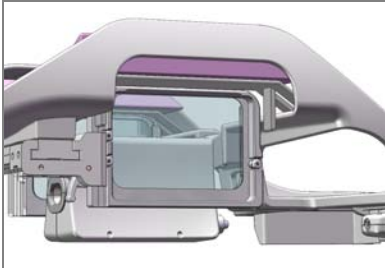


Figure 45: Correct Hand Positioning

The following table describes how to position the shoulder supports and pads.

Table 9: Shoulder Support Placement

Image	Instruction
	Step 1. Attach shoulder support pad to shoulder support. Position immobilization frame approximately 50mm (two inches) from arch.
	Step 2. Place shoulder support with shoulder support pad between immobilization frame and arch. The immobilization frame should fit under the medial side of the support. Make sure the vertical tabs are positioned on the inside of the gap to provide added support.

Image	Instruction
	Step 3. Repeat placement procedure for opposite shoulder support. The supports should be placed as far laterally as possible before the patient is placed on the tabletop. The body wedge pad should be pulled up to meet the shoulder support pads ensuring there is no gap between the body wedge pad and the edge of the patient support.
	<i>NOTE: Make sure the shoulder support stays in place during the procedure. This will help relieve pressure on the patient's sternum.</i>

Arms-Back Positioning

When positioning patients with their arms back:

- Ensure their arms are by their sides.
- Remove the front pad for better air flow. The front pad is optional for arms-back positioning.
- Use shoulder support bridges and pads to support torso weight and release pressure from the sternum. The shoulder support bridges and pads can be added after positioning the coils.
- Use the hip wedge pad. Raising the hips flattens the back and relieves pressure on the sternum. Placing the hip wedge pad under the body wedge pad may increase comfort for some patients.
- Adjust the headrest height – the head should be aligned with spine to help release pressure from the sternum.
- Use foot rest pad. Position the foot rest pad with the edge aligned with the patient's knees, supporting the feet.

Arms-Forward Positioning

When positioning patients with their arms forward:



- If the patient is in the arms-forward position, the hand separation pad must be used between patient's hands so that a loop is not created.

- Use the front pad to comfort the arms. Note that the patient's arms might become numb with arms-forward positioning.
- Remove the shoulder support pads.
- Use the hip wedge pad. Raising the hips flattens the back and relieves pressure on the sternum. Placing the hip wedge pad under the body wedge pad may increase comfort for some patients.
- Use foot rest pad. Position the foot rest pad with the edge aligned with the patient's knees, supporting the feet.

IV Management

Intravenous protocols differ from site to site. Follow the protocol established at your facility.

- For unilateral interventional procedures, the IV is ideally inserted into the arm opposite to the side of the biopsy. This arm can be positioned over the patient's head for easy access during the procedure.
- Have the injection pump ready before the patient enters the exam room.
- While patient is being positioned on the Tabletop, attach the extension tubing to IV.
- With the patient lying with their arms back, loop the extension tubing over patient's thumb. Extend the tubing over the patient's back and secure the position with tape i.e. you can place a piece of tape over the patient's shoulder.
- Flush IV with saline to ensure proper position. Feel the flush go through the arm. Ask the patient if there is any pain.
- As the patient is shuttled into scanner, ensure the extension tubing and other pieces of equipment lie within the patient support and never outside the patient support arches.



To avoid possible patient injury, do not use the automatic button to advance the patient into the scanner. Use the manual advance button and watch for potential contact of the patient with the bore as they advance into the scanner.

- Position the injector close to the MR scanner.

Chapter 4 Imaging with the Tabletop

This chapter outlines how to set up and use the Tabletop for bilateral breast imaging. Use the same procedure, adjusted appropriately, for unilateral breast imaging.

Safety Precautions



Do not handle the electrical pins in the connector housings of the system plug, the coil connectors in the cable tray, or the coils. Electrostatic discharge could damage the equipment.



CAUTION

To avoid possible patient injury, do not use the automatic button to advance the patient into the scanner. Use the manual advance button and watch for potential contact of the patient with the bore as they advance into the scanner.



Keep limbs away from the gap between the Tabletop and the scanner while raising and lowering the MR patient table.



Patients must not form a loop with any body parts while scanning. For example, they cannot touch any part of the right hand or arm to any part of the left hand or arm. This could cause a burn at the point of contact. Use pads to prevent any part of the patient's body from touching metal on the patient table. Educate the patient accordingly and check the patient's position immediately before the scan.

Imaging Procedure

Checklist

- 2 Lateral Array Coils, or 2 Single Loop Coils for 10-channel imaging if that configuration is available with your system
- 1 Medial Array Coil
- 2 Clear Immobilization Plates (if using Single Loop Coils)
- Padding

System Setup and Patient Positioning

1. Adjust the Tabletop padding to accommodate for patient size and comfort. Larger patients may require less padding. See the Patient Positioning and Preparation Guidelines Chapter, beginning on page 45 for details.
2. Wipe down all padding with an approved disinfectant (see Cleaning on page 64) and lay clean linens on all surfaces that may contact the patient.
3. Open the coils or immobilization frames to their most lateral position.
4. Insert the appropriate coils. See Inserting and Connecting the Coils on page 31 for more information.
5. Open the cover of the interface box under the ramp pad and connect the lateral array coils (left and right) and medial array coil to the connection ports.
6. Plug the Tabletop cable into the correct scanner port.
7. Ensure the patient is wearing appropriate hearing protection.
8. Lower the 3.0T GE MRI System table.
9. Assist the patient onto the Tabletop. Patients can steady themselves with the patient support arches as they lower their breasts into the aperture.
10. Raise the 3.0T GE MRI System Table for easier patient positioning.
11. Position the patient's arms as per the guidelines in the Patient Positioning and Preparation Guidelines Chapter, beginning on page 45.
12. Ensure the patient is aligned centrally on the patient support.
13. To ensure the patient's breasts are centered with respect to the coils, remove the sliders from the slider rail, and observe the position of each breast. If necessary, ask the patient to adjust their position until the breasts are properly positioned.
14. Pull the breast tissue down and away from the chest wall ensuring no tissue is caught on the posterior side of the immobilization plate or sternum support.
15. Smooth breast tissue on the lateral side with thumb to remove mammary fold.
16. Replace the slider on that side.

NOTE: Be careful not to pinch the patient when replacing the slider.

17. With the coil set at optimal height, release the horizontal lock on the slider and move it medially to offer immobilization but not compression of the breast. Lock the slider. Repeat positioning for the contralateral breast.

NOTE: The figures below show the immobilization frame and lateral coil set at optimal height, approximately 25 mm (one inch) above the height of the medial coil array.

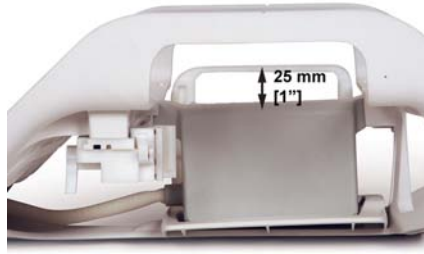


Figure 46: Immobilization Frame Positioned at Optimal Height

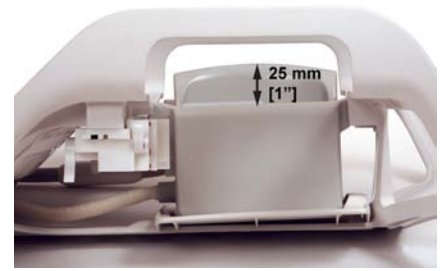


Figure 47: Lateral Coil Positioned at Optimal Height

18. Adjust the headrest as needed.
19. Relieve pressure on the patient's sternum by positioning the hip wedge pad to raise the hips and flatten the back. If a hip wedge pad is not available, a pillow in the same position will suffice.
20. Place the foot rest pad under the patient's legs to improve comfort.
21. Advance the patient into the MR scanner, landmark on the centre of the coils and advance patient support to scan.
22. Proceed with the imaging protocol.

Post-Examination Procedures

1. Shuttle the patient out of the MR scanner.
2. Release the immobilization frames.
3. Lower the 3.0T GE MRI Systems table and assist the patient off the Tabletop.
4. Clean all Tabletop surfaces following the instructions in the Maintenance Chapter, beginning on page 63.

Chapter 5 Interventional Procedures with the Tabletop

This chapter describes how to set up and use the Tabletop for unilateral and bilateral interventional procedures.

NOTE: This chapter is not applicable to a 16-channel imaging-only system.

Always review the patient's previous imaging to find where the target is before the exam. This allows you to set up the Tabletop properly and optimize patient positioning.

Safety Precautions



Do not handle the electrical pins in the connector housings of the system plug, the coil connectors in the cable tray, or the coils. Electrostatic discharge could damage the equipment.



To avoid possible patient injury, do not use the automatic button to advance the patient into the scanner. Use the manual advance button and watch for potential contact of the patient with the bore as they advance into the scanner.



Keep limbs away from the gap between the Tabletop and the scanner while raising and lowering the MR patient table.



Patients must not form a loop with any body parts while scanning. For example, they cannot touch any part of the right hand or arm to any part of the left hand or arm. This could cause a burn at the point of contact. Use pads to prevent any part of the patient's body from touching metal on the patient table. Educate the patient accordingly and check the patient's position immediately before the scan.

Safety Notices for Interventional Procedures

The following safety notices apply specifically to interventional procedures with the Tabletop.



Performing interventional procedures on female patients with breast implants may cause the implant to rupture.

If a biopsy has previously been performed, do not use MR reference images created before the biopsy. Create new MR images in case the anatomy changed as a result of the biopsy.

Perform a biopsy only when both the marker and lesion are located within an ellipsoid with the following axial expansions about the magnet iso-centre: x (right/left) = 44 cm, y (anterior/posterior) = 26 cm, z (head/feet) = 26 cm. Objects outside this area may not be accurately represented in images due to the presence of significant field inhomogeneities, which could result in the biopsy missing the lesion.

To select the reference position, select only slices that run parallel to the grid (sagittal slices). Measure the slices without slice distance and a slice thickness less than or equal to 2 mm.

Select the tip of the marker to compute the reference position.



Use a needle block that corresponds to the diameter of the needle thickness.

Use biopsy needles whose lengths correspond to the calculated puncture depth.

To correctly position the needle, take the length of the needle block into account. Consult the documentation provided by the needle vendor regarding selecting the correct needle depth.

Equipment Checklist

Unilateral Intervention

- 1 Contralateral Breast Support
- 1 Right Single Loop Coil
- 1 Left Single Loop Coil
- Medial Plug
- 2 Biopsy Grids
- MRI Biopsy Worksheet (Provided by Sentinelle Medical or your needle vendor)

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

Bilateral Intervention

- 1 Right Single Loop Coil
- 1 Left Single Loop Coil
- 1 Medial Array Coil

- 2 Biopsy Grids
- MRI Biopsy Worksheet (Provided by Sentinelle Medical or your needle vendor)

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

Biopsy Supplies Not Included (Recommended)

The following list is an example of supplies that may be required when performing biopsies. These supplies are not provided by Sentinelle Medical. Discuss with the respective needle vendor specific supplies that are required for the vacuum assisted device prior to the biopsy date.

Sterile Tray

- Sterile drape on which to place sterile supplies
- Sterile gloves
- Povidone-iodine topical antiseptic swabs
- Local anaesthetic
- Local anaesthetic with Epinephrine
- 3cc Syringe (1)
- 10cc Syringe (2)
- 18G 1½" needle (to draw anaesthetic)
- 25G 1½" needle (for anaesthetic prior to skin nick)
- 22G 3" needle (for anaesthetic administration w/deeper lesion)
- Scalpel (#11)
- 4x4" gauze
- 2x2" gauze
- Thumb forceps

Disposables Non-Sterile

- Chucks, towels, or absorbent non-sterile material
- Auto or Manual Fiducial markers (2 Auto Fiducials or up to 6 Manual Fiducials for bilateral interventional procedures, or 1 Auto Fiducial or up to 3 Manual Fiducials for unilateral interventional procedures), or vitamin E capsules
- Non-sterile gloves

Post Procedure Prep Checklist

- Large Formalin Container (for specimen)
- Pathology information/request form
- Gauze for post biopsy-compression
- Steri-strips™ to close incision
- Medical dressing to cover steri-strips™ (optional)
- Cold pack
- Post-procedure information sheet
- Large sharps container

Other Required Equipment Not Supplied by Sentinelle Medical

- Absorbent Pads
- Post Operative Care Sheet
- Biohazard Waste Containers

Interventional Procedure

Frames and Coils Setup for Bilateral Procedures

Follow these steps to set up the Vanguard for bilateral interventional procedures. See the next section for setup instructions for unilateral procedures.

1. Unplug and remove the lateral array coils.
2. Insert the immobilization frames and biopsy grids.



Figure 48: Immobilization Frame in Slider

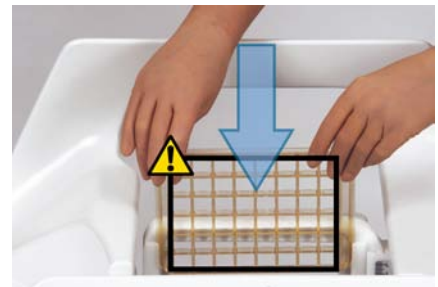


Figure 49: Inserting a Biopsy Grid



CAUTION

Always follow your clinical gloving procedures to ensure that the sterility of the grid surfaces is not compromised.

NOTE: The sterile biopsy grid is a single-use device. Reusing this device risks patient infection and/or the spread of pathogens.

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

3. Position the immobilization frames to their most lateral position.
4. Install the single loop coils in the uppermost position in the immobilization frames, and connect them. If the top or bottom row of the grid is blocked by the single loop coil, move the single loop coil anteriorly/posteriorly in the immobilization frame to allow access.
5. If the medial coil is not already in place, insert it and connect it.

Frames and Coils Setup for Unilateral Procedures

Follow these steps to set up the Vanguard for unilateral interventional procedures. See the previous section for setup instructions for bilateral procedures.

1. Remove the medial and lateral array coils. Store them in a drawer.
2. Insert the immobilization frames and biopsy grids.



Figure 50: Immobilization Frame in Slider

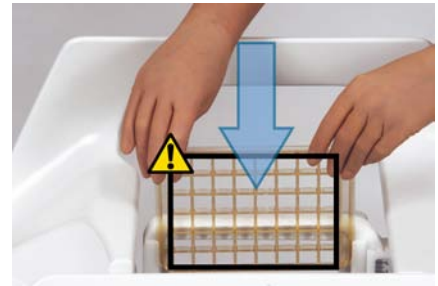


Figure 51: Inserting a Biopsy Grid



CAUTION

Always follow your clinical gloving procedures to ensure that the sterility of the grid surfaces is not compromised.

NOTE: The sterile biopsy grid is a single-use device. Reusing this device risks patient infection and/or the spread of pathogens.

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

3. Unlock the horizontal locks of the immobilization frames and slide both frames to the side of biopsy (right for right breast or left for left breast).
4. Place the contralateral breast support to support the opposite breast during the procedure. Ensure the support is angled down lateral to medial.
5. Position the medial immobilization frame beside and at the same height as the contralateral breast support. Lower slightly and lock in place.
6. Position the lateral immobilization frame to its most lateral position.

7. Install the single loop coils in the uppermost position in the immobilization frames, and connect them. If the top or bottom row of the grid is blocked by the single loop coil, move the single loop coil anteriorly/posteriorly in the immobilization frame to allow access.
8. Plug the medial plug into the medial connector.

Patient Positioning

1. Adjust the Tabletop padding to accommodate for patient size and comfort. Larger patients may require less padding. See the Patient Positioning and Preparation Guidelines Chapter, beginning on page 45 for details.
2. Wipe down all padding with an approved disinfectant (see Cleaning on page 64) and lay clean linens on all surfaces that may contact the patient. and lay clean linens and absorbent pads on the table surface and all surfaces that may contact the patient, including the Contralateral Breast Support if present.
3. Clean and prepare the patient's breast according to site policies and procedures.
4. Ensure the patient is wearing appropriate hearing protection.
5. Lower the 3.0T GE MRI System table.
6. Assist the patient onto the Tabletop. Patients can steady themselves with the patient support arches as they lower their breasts into the aperture.

For unilateral procedures, the target breast will be between the two biopsy grids and the opposite breast will rest comfortably on the contralateral breast support.

For bilateral procedures, each breast will be between a biopsy grid and the medial array coil.
7. Raise the 3.0T GE MRI System table for easier patient positioning.
8. Position the patient's arms as per the guidelines in the Patient Positioning and Preparation Guidelines Chapter, beginning on page 45.
9. Ensure the patient is aligned centrally on the patient support.
10. Have the patient adjust their position as necessary to ensure that the region of interest is centered with respect to the biopsy grid.
11. Adjust the headrest as needed.
12. Relieve pressure on the patient's sternum by positioning the hip wedge pad to raise the hips and flatten the back. If a hip wedge pad is not available, a pillow in the same position will suffice.
13. Pull the breast tissue down and away from the chest wall, ensuring no tissue is caught on the sternum support.

14. (Unilateral procedures only) Apply light immobilization by moving the medial frame. Once positioned, ensure the medial frame is locked both horizontally and vertically.
15. Smooth breast tissue on the lateral side with thumb to remove mammary fold.
16. Apply sufficient immobilization with the lateral grid to keep the breast in position. The breast should feel like the palm of your hand when fingers are extended.
17. If you are using a Sentinelle Auto Fiducial, place it in grid window C4. Otherwise, place your fiducials or vitamin E capsules in grid windows in close proximity to the target. Document these locations on the appropriate grid worksheet (provided by Sentinelle Medical or your needle vendor).

NOTE: Sentinelle Auto Fiducials are for use with Sentinelle Medical Aegis software only. If you are not using Aegis to analyze your MR images, you must use manual fiducials or vitamin E capsules.

18. For bilateral procedures, repeat this process (positioning the breast, applying immobilization with the biopsy grid, and placing fiducials in the appropriate grid window) for the other breast.



Figure 52: Pulling Breast Tissue Down (Grid Removed to Show Detail)



Figure 53: Automatic Fiducial Marker in Grid

19. Place the foot rest pad under the patient's legs to improve comfort.
20. Plug the Tabletop cable into the correct scanner port.
21. Advance the patient into the MR scanner, landmark on the centre of the coils and advance patient support to scan.
22. Proceed with the imaging protocol.

Post-Examination Procedures

1. Shuttle the patient out of the MR scanner.
2. Release the immobilization frames.
3. Treat the biopsy site according to your site's recommendations.

4. Assist the patient off the Tabletop.
5. Dispose of the grids in a biohazard container.
6. Clean all Tabletop surfaces following the instructions in the Maintenance Chapter, beginning on page 63.

Recommended Scan Protocols

Any specific choice of protocols is the responsibility of the radiologist and/or technologist. A set of protocols commonly used for breast exams at MR sites can be provided by contacting Sentinelle Medical.

Chapter 6 Maintenance and Troubleshooting

Storage

Coils and apparatus should be stored at the same room temperature and relative humidity as the MR scanner. Storing the Tabletop in the MR Suite and scanner room is acceptable.

When not in the scanner room, transport and store the Tabletop and all components under the conditions described in the following table.

Table 10: Transportation and Storage Environment

	Minimum	Maximum
Ambient Temperature (°C)	-30	+65
Relative Humidity (non-condensing)	10%	95%
Atmospheric Pressure	500 hPa	1060 hPa

Inspection

Inspect the coils weekly for signs of mechanical damage or breakage. Inspect to see that the system fits securely in the MR scanner patient table and that the patient table motion into the scanner is smooth and regular.



CAUTION

Do not use the Tabletop if it has sustained mechanical damage or if patient support motion is significantly compromised. Contact a Sentinelle Medical representative immediately for the best course of action.

It is important to inspect the coils to ensure the coil enclosure is not cracked or otherwise compromised in such a way that it may no longer be watertight. If you suspect a coil may be damaged, contact a Sentinelle Medical service representative. For contact information, see Contacting Sentinelle Medical on page 13.

Special Care Requirements

Cleaning

The Sentinelle Medical biopsy grids are sterile, single-use disposable items. The following parts, however, should be cleaned before and after each use.

NOTE: Use protective gloves to perform these cleaning tasks, and always comply with your site's biohazard/blood-borne pathogen safety protocols.

Routine Imaging:

- The Tabletop Immobilization System
- Padding
- Medial Array Coil
- Left and Right Lateral Array Coils

Interventional Imaging:

- The Tabletop Immobilization System
- Padding
- Fiducials
- Left and Right Single Loop Coils
- Medial Array Coil (for bilateral interventions)
- Contralateral Breast Support (for unilateral interventions)
- Catchment Tray

The cleaning solutions listed in the table below have been tested and are recommended for cleaning the apparatus and the coils. Use a cotton cloth to clean the coils, system surfaces, and padding.



CAUTION

The coils are not waterproof and should not be immersed in any liquid. This will damage the coils and may cause injury.

Do not use cleaning solutions not listed in this table as they may damage the system. Sentinelle Medical must approve any proposed cleaning solutions.

NOTE: To avoid soiling the coils, place a cotton sheet over the coil surfaces before positioning the patient. If a coil becomes soiled, clean it as described.

Table 11: Approved Cleaning Solutions

Solution	Comments
Warm water	Safe for all areas.
Commercial dish washing liquid/water combination	Safe for all areas.

Solution	Comments
Alcohol solution (70%, isopropyl alcohol, 30% water).	Safe for patient support surfaces, coil housings, and immobilization plates. Do not apply to Padding surfaces. Do not apply to adhesive-backed materials such as labels and Velcro® fasteners.
Hydrogen Peroxide-based cleaners (e.g., Virox, G-Force® H2O2 and Accel TB™)	Safe for patient support surfaces, coil housings, and immobilization plates. Do not apply to Padding surfaces. Do not apply to adhesive-backed materials such as labels and Velcro® fasteners.
Anticeptizyme	Safe for use on immobilization frames, patient support, and coil housings to break up bodily fluids as needed.
Virex TB	Safe for Padding.
Heptagon II Disinfectant Spray	Safe for Padding.
VC 79	Safe for Padding.
1:10 dilution of household bleaches containing 5.25% sodium hypochlorite	Safe for Padding.

NOTES:

- Do not apply tape to the top surface of the padding.
- Do not mark the padding with ink.
- Do not immerse the coil cable in cleaning fluid. Isopropyl alcohol can be used to wipe the contacts as required.
- If bodily fluids contact immobilization frames or coil enclosures, use the appropriate decontamination techniques to clean them.
- Clean all waste from the coil using 1:10 bleach/water solution and a cloth.
- Ensure coil enclosure is intact and is not permeable to fluid.
- Coils cannot be sterilized using any autoclaving techniques.

Replaceable Components/Accessories

Contact your service representative for replacement parts. For contact information, see Coil Service on page 34.

Troubleshooting

This section lists troubleshooting suggestions for common scanning problems. If you cannot resolve a problem after trying these solutions, contact your service representative to arrange for service or repairs to the system. See below.

Table 12: Troubleshooting

Problem	Possible Solution(s)
You are unable to pre-scan or are scanning and receiving no signal.	Verify that: • You have selected the correct coil configuration. Refer to Chapter 1 to verify the proper coils have been selected. • The coils are properly attached to the immobilization frames, properly connected to the signal lines, and correspond to the coil configuration. • The landmark is correct. • The scan locations and any FOV offsets are correct. <i>NOTE: When selecting two separate 2D imaging volumes, the scan may select the middle slice of the entire set of slices for prescan. Ensure the middle slice is covering an appropriate volume of interest.</i> <i>If a scan still cannot be performed, remove all of the coils, terminate all of the signal lines, and attempt to scan using the body coil and the identical imaging sequence and parameters. If there is still no signal received, the problem is probably with the MRI system. Please contact your service representative.</i>
Signal quality obtained in the quality assessment check is not acceptable or clinical imaging scan quality is not acceptable.	• Ensure the protocol used in the periodic quality assessment check is identical to initial baseline scan. • Verify that all coils are properly selected. • Verify that coils are in proper positions. • Verify that there are no loops in the cables. • Verify that the imaging area is free from ferromagnetic and foreign objects. • Verify that the center frequency is within the adjustment range of your system. • Ensure electrical contacts on the coils and the MRI connectors are clean and free from obstruction. If they are not clean, they can be cleaned with a cotton swab and isopropyl alcohol.
A dark region appears in the image.	Verify that there are no ferromagnetic or foreign objects within the imaging volume.

Appendix A

NOTE: Items on the following form are only available directly from Sentinelle Medical.

Sentinel Medical, a Division of Hologic
Confidence in Diagnosis
Accuracy in Intervention

Order Form

555 Richmond St. W, Suite 800, P.O. Box 301
 Toronto, ON M5V 3B1

Phone: (North America) 1-866-735-3744
 (International) 1-416-366-4994

Fax: 1-647-258-0299

The following number must appear on all related correspondence,
 shipping papers, and invoices:

P.O. NUMBER:

Purchaser

Name:
 Company:
 Address:
 City, State/Province:
 ZIP/Postal Code:
 Phone:
 Fax:

Ship To

Name:
 Company:
 Address:
 City, State/Province:
 ZIP/Postal Code:
 Phone:
 Fax:

P.O. DATE

QTY	Part Number	DESCRIPTION	UNIT PRICE	TOTAL
	5000137-51	Box of 10 Sterile Biopsy Grids for use with the Tabletop	\$400.00*	
	SM-0517	Box of 25 Disposable Head Rest Covers	\$150.00*	
	SM-0511	Box of 25 Disposable Table Covers	\$150.00*	
*Please confirm price with sales dept.		SUBTOTAL		
		SALES TAX (GST IN CANADA)		
		SHIPPING & HANDLING		
		OTHER		
Send all correspondence to:		TOTAL		

Sales Dept.
 555 Richmond St. W
 Suite 800, P.O. Box 301
 Toronto, ON M5V 3B1

Authorized by

Date

